Highlights from the three-spring problem: the same problem, solved two ways

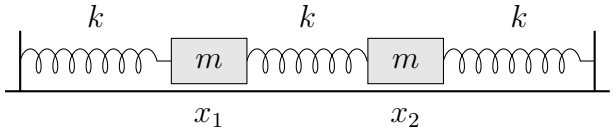
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These notes closely follow Felder and Felder, Section 6.9, "Putting It Together: Revisiting the Three-Spring Problem"[1] (pp. 336–340), sometimes to the point of plagiarism, so I'll just give full credit now. The book works the problem on a matrix with eigenvalues -4 and -1; here I've swapped in the carts we actually have on the track.

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1 The problem



Question: Suppose we have two carts on a smooth track (perhaps an airtrack), each with mass m, and connected by three equal springs with spring constants k. Find the solution (i.e. $x_1(t), x_2(t)$) with initial conditions $x_1(0) = 3$ cm, $x_2(0) = 0$, $\dot{x}_1(0) = \dot{x}_2(0) = 0$. Here x_1 and x_2 are each cart's displacement from its own equilibrium position, with positive x to the right.

Step 1: draw the picture, write down N2L. Cart 1 feels the left spring, stretched by x_1, pulling it back, and the middle spring, stretched by $x_2 - x_1$, pulling it forward:

$$m\ddot{x}_1 = -kx_1 + k(x_2 - x_1) = -2kx_1 + kx_2 \tag{1.1}$$

$$m\ddot{x}_2 = -2kx_2 + kx_1 \tag{1.2}$$

(We get the second line by symmetry, just swapping the labels; conceptually the middle spring pushes the two carts in opposite directions, which is why the off-diagonal terms below are both positive while the diagonal ones are both negative.)

Step 2: turn it into a matrix equation. Writing $\omega_0^2 \equiv k/m$,

$$\begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = \omega_0^2 \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad \text{i.e.} \qquad \ddot{\vec{x}} = \mathbf{M}\vec{x}. \tag{1.3}$$

The column is the *state* of the system (where both carts are), and $\mathbf{M}$ is a transformation acting on it. Intuitively, equations (1.1) and (1.2) just *look* like a matrix. More formally, we note that the

state we are transforming needs more than one number, and each derivative is a linear combination of those numbers.

Step 3: find the eigenvalues and eigenvectors of M. Both methods below start here, so let's get it out of the way. By definition $\mathbf{M}\vec{v} = \lambda\vec{v}$, so if we plug in $\vec{v}$, the right-hand side will simplify greatly. Let's see it in action. To find λ we need $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$. We talked extensively in class about how this relates to linear dependence. We'll find λ and $\vec{v}$ for $\left(\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$, knowing we can multiply by ω_0^2 at the end:

$$\begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} = (-2-\lambda)^2 - 1 = \lambda^2 + 4\lambda + 3 = (\lambda+1)(\lambda+3) = 0 \quad \Rightarrow \quad \lambda \in \{-1, -3\}. \tag{1.4}$$

Now find the eigenvectors: $(\mathbf{M} - \lambda\mathbf{I})\vec{v} = 0$ says

$$(-2-\lambda)x_1 + x_2 = 0 \tag{1.5}$$

(the second row says the same thing; if it didn't, you'd have the wrong λ). Plug $\lambda = -1$ into (1.5): $-x_1 + x_2 = 0$, so $x_2 = x_1$, or $\vec{v}_1 = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$. Any multiple of an eigenvector is an eigenvector, so we could have picked $\left(\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right)$, etc. Plug in $\lambda = -3$: $x_1 + x_2 = 0$, so $\vec{v}_2 = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. Putting the ω_0^2 back,

$$\boxed{\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix},\ \lambda_1 = -\omega_0^2; \qquad \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},\ \lambda_2 = -3\omega_0^2.} \tag{1.6}$$

When we interpret this, we say "the eigenvalue is $-\omega^2$." You might be tempted to over-interpret and say "the frequency is the square root of the absolute value of the eigenvalue" but that's an over-generalization. The minus sign is the physics: a negative eigenvalue means oscillation, a positive one would mean exponential growth.

2 The first way: plug in one eigenvector at a time

$$\boxed{\vec{v}_1 = \begin{pmatrix}1\\1\end{pmatrix},\ \lambda_1 = -\omega_0^2; \qquad \vec{v}_2 = \begin{pmatrix}1\\-1\end{pmatrix},\ \lambda_2 = -3\omega_0^2.}$$

Each eigenvector becomes a normal mode. The first eigenvector tells us that if you plug in *any* state in which $x_2 = x_1$, the matrix will simply multiply that state by $-\omega_0^2$. Now, let's plug in. That means $x_1 = x_2$, so Equation (1.3) becomes

$$\begin{pmatrix}\ddot{x}_1\\ \ddot{x}_2\end{pmatrix} = \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} \quad \rightarrow \quad \begin{pmatrix}\ddot{x}_1\\ \ddot{x}_1\end{pmatrix} = \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_1\end{pmatrix}. \tag{2.1}$$

And now we use $\mathbf{M}\vec{v} = \lambda\vec{v}$: the column on the right is a multiple of the eigenvector $\vec{v}_1$, so $\mathbf{M}$ acting on it is just $\lambda_1 = -\omega_0^2$ times it, and we get

$$\begin{aligned} \begin{pmatrix}\ddot{x}_1\\ \ddot{x}_1\end{pmatrix} &= \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_1\end{pmatrix} \\ &= \mathbf{M}\ \vec{v} \\ &= \lambda\ \vec{v} \\ \rightarrow \quad \ddot{x}_1 &= -\omega_0^2\ x_1 \end{aligned} \tag{2.2}$$

(In fact we have this same equation $\ddot{x}_1 = -\omega_0^2 x_1$ twice.) The matrix multiplication became a scalar multiplication. That we know how to solve, and the solution is the first normal mode:

$$x_1(t) = A_1 \cos\omega_0 t + A_2 \sin\omega_0 t, \qquad x_2(t) = x_1(t). \tag{2.3}$$

Now the other eigenvector, $x_2 = -x_1$. Written out as a matrix equation (even though we only need the column),

$$\begin{aligned} \begin{pmatrix}\ddot{x}_1\\ -\ddot{x}_1\end{pmatrix} = -3\omega_0^2 \begin{pmatrix}x_1\\ -x_1\end{pmatrix} \quad &\rightarrow \quad \ddot{x}_1 = -3\omega_0^2 x_1 \\ &\rightarrow \quad x_1(t) = B_1 \cos\sqrt{3}\,\omega_0 t + B_2 \sin\sqrt{3}\,\omega_0 t, \qquad x_2(t) = -x_1(t). \end{aligned} \tag{2.4}$$

Cool! That's the $\sqrt{3}$ we pulled out of a *hat* back when we first met these carts.[1]

The general solution is the sum of the two normal modes. Two second-order equations, four constants. Now we apply our initial conditions. Since $\dot{x}_1(0) = \dot{x}_2(0) = 0$, we know that we discard the sine terms, and we're left with

$$\begin{aligned} x_1(t) &= A_1 \cos\omega_0 t + B_1 \cos\sqrt{3}\,\omega_0 t \\ x_2(t) &= A_1 \cos\omega_0 t - B_1 \cos\sqrt{3}\,\omega_0 t. \end{aligned} \tag{2.5}$$

Let's now be a little clever!

$$\begin{array}{c} \\ x_1(0) \\ x_2(0) \end{array} \begin{array}{c} \begin{array}{cc} A_1 & B_1 \end{array} \\ \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{array} \tag{2.6}$$

[1] Try replacing the state with something that isn't a normal mode, say $x_2 = 3x_1$: the two rows give you two contradictory differential equations. Only the eigenvectors work!

is the matrix that converts normal-mode amplitudes into initial positions. We can tell that from the column labels, and apparently it's a matrix of eigenvectors. Why? Well,

$$\begin{aligned} x_1 &= \overbrace{A_1 \cos\omega_0 t}^{\text{normal mode 1}} \quad + \overbrace{B_1 \cos\sqrt{3}\,\omega_0 t}^{\text{normal mode 2}} \\ x_2 &= A_1 \cos\omega_0 t \quad - B_1 \cos\sqrt{3}\,\omega_0 t \end{aligned} \tag{2.7}$$

Now plug in $t = 0$. The cosines are all 1, and we get

$$\begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} A_1 + B_1 \\ A_1 - B_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix}. \tag{2.8}$$

Each column is one normal mode at $t = 0$, which is an eigenvector. Cool. So, the matrix that converts normal-mode amplitudes to positions is the matrix of eigenvectors, and the matrix that converts positions to amplitudes is its inverse, with the labels swapped:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{-2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{matrix} \\ A_1 \\ B_1 \end{matrix} \begin{matrix} \begin{matrix} x_1(0) & x_2(0) \end{matrix} \\ \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \end{matrix}. \tag{2.9}$$

So, that last matrix converts initial conditions to normal-mode amplitudes. Applying it to ours,

$$\begin{pmatrix} A_1 \\ B_1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}, \tag{2.10}$$

so the solution is

$$\boxed{x_1(t) = \tfrac{3}{2}\cos\omega_0 t + \tfrac{3}{2}\cos\sqrt{3}\,\omega_0 t, \qquad x_2(t) = \tfrac{3}{2}\cos\omega_0 t - \tfrac{3}{2}\cos\sqrt{3}\,\omega_0 t} \tag{2.11}$$

in cm, with $\omega_0 = \sqrt{k/m}$. Whew!

Check it at $t = 0$: $x_1 = 3$, $x_2 = 0$. Good. Pulling one cart and holding the other at rest is half of each normal mode, so each cart carries both frequencies, ω_0 and $\sqrt{3}\,\omega_0$. Their periods are in the ratio $\sqrt{3}$, whatever k and m are. And since two cosines with frequencies in an irrational ratio never line up again, the motion never exactly repeats. That's the "janky" run we saw on the track.

One caution. The matrix in (2.6) is the matrix of eigenvectors *only at* $t = 0$. At a general t the matrix between (A_1, B_1) and (x_1, x_2) is

$$\begin{matrix} \\ x_1(t) \\ x_2(t) \end{matrix} \begin{matrix} \begin{matrix} A_1 & B_1 \end{matrix} \\ \begin{pmatrix} \cos\omega_0 t & \cos\sqrt{3}\,\omega_0 t \\ \cos\omega_0 t & -\cos\sqrt{3}\,\omega_0 t \end{pmatrix} \end{matrix}.$$

Setting $t = 0$ is what turned it into $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$.

3 The "better" way: the natural basis

$$\boxed{\vec{v}_1 = \begin{pmatrix}1\\1\end{pmatrix},\ \lambda_1 = -\omega_0^2; \qquad \vec{v}_2 = \begin{pmatrix}1\\-1\end{pmatrix},\ \lambda_2 = -3\omega_0^2.}$$

This way gets the same answer, but it shows what "basis" and "diagonalization" mean more directly. Above, we plugged the eigenvectors in one at a time and watched what they did to the equation. Instead, let's rewrite the *entire* equation in the basis defined by the eigenvectors.

Question: Find the general solution to the same equations, the carts with equal masses m,

$$\begin{pmatrix}\ddot{x}_1\\ \ddot{x}_2\end{pmatrix} = \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}.$$

Answer: The eigenvectors of this transformation matrix are $\left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)$, with corresponding eigenvalues $-\omega_0^2$ and $-3\omega_0^2$ (boxed above). So, we can write x_1 and x_2 in this basis as

$$\begin{pmatrix}x_1(t)\\ x_2(t)\end{pmatrix} = f(t)\begin{pmatrix}1\\1\end{pmatrix} + g(t)\begin{pmatrix}1\\-1\end{pmatrix}. \tag{3.1}$$

This says "you can express the vector (x_1, x_2) at any given time as a linear combination of these two basis vectors, with f and g as the coefficients": f is how much of the first normal mode we have, g how much of the second. This changes in time, so f and g are functions of t. Instead of asking "where is each cart?" we are asking "how much of each normal mode do we have?"

Can we do that? Yes, for absolutely any functions x_1 and x_2, and not because $\vec{v}_1$ and $\vec{v}_2$ are eigenvectors: any two non-parallel vectors form a basis, so any vector is some combination of them. That's Section 6.4.

What happens when we do that? When we express all our states in a different basis, the entire equation changes, including the transformation matrix itself. We'd like to write a corresponding equation to (1.3) in this basis:

$$\begin{pmatrix}\ddot{f}\\ \ddot{g}\end{pmatrix} = \begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}f\\ g\end{pmatrix}, \tag{3.2}$$

and we need to know what a, b, c, d are. Two key points from linear algebra:

1. When you use the eigenvectors as a basis, the transformation is represented by a diagonal matrix of the eigenvalues. Let's write this out! Call the eigenvectors $\vec{v}_1$, $\vec{v}_2$ with corresponding eigenvalues λ_1, λ_2, and take any vector $\vec{v} = a\vec{v}_1 + b\vec{v}_2$. Apply $\mathbf{T}$:

   $$\mathbf{T}\vec{v} = \mathbf{T}(a\vec{v}_1 + b\vec{v}_2) = a\,\mathbf{T}\vec{v}_1 + b\,\mathbf{T}\vec{v}_2 = a\lambda_1\vec{v}_1 + b\lambda_2\vec{v}_2.$$

   Now read that in the $\vec{v}_1\vec{v}_2$ basis. The input coordinates are $\left(\begin{smallmatrix}a\\b\end{smallmatrix}\right)$. The output is $\left(\begin{smallmatrix}\lambda_1 a\\ \lambda_2 b\end{smallmatrix}\right)$. So

   $$\mathbf{T} = \begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix}$$

   in this basis. (§4 says the same thing more slowly.)

2. This naturally decouples our equations.

We can immediately write it. In the eigenvector basis, (1.3) becomes

$$\begin{matrix} & f & g \\ \ddot{f} & & \\ \ddot{g} & & \end{matrix} \begin{pmatrix} \ddot{f} \\ \ddot{g} \end{pmatrix} = \begin{pmatrix} -\omega_0^2 & 0 \\ 0 & -3\omega_0^2 \end{pmatrix} \begin{pmatrix} f \\ g \end{pmatrix} \qquad \Rightarrow \qquad \ddot{f} = -\omega_0^2 f, \quad \ddot{g} = -3\omega_0^2 g. \tag{3.3}$$

The natural basis has done its job. Those are two separate simple harmonic oscillators, so

$$f = A_1 \cos\omega_0 t + A_2 \sin\omega_0 t, \qquad g = B_1 \cos\sqrt{3}\,\omega_0 t + B_2 \sin\sqrt{3}\,\omega_0 t. \tag{3.4}$$

Finally, we go back to our position basis (x_1, x_2) with (3.1), $x_1 = f + g$ and $x_2 = f - g$:

$$\boxed{\begin{aligned} x_1 &= f + g = A_1 \cos\omega_0 t + A_2 \sin\omega_0 t + B_1 \cos\sqrt{3}\,\omega_0 t + B_2 \sin\sqrt{3}\,\omega_0 t \\ x_2 &= f - g = A_1 \cos\omega_0 t + A_2 \sin\omega_0 t - B_1 \cos\sqrt{3}\,\omega_0 t - B_2 \sin\sqrt{3}\,\omega_0 t \end{aligned}} \tag{3.5}$$

So, four constants, as there must be. Same answer as (2.5) once the sines are gone.

The initial conditions, this way. Solving (3.1) for the coefficients gives $f = (x_1 + x_2)/2$ and $g = (x_1 - x_2)/2$ (that's the inverse matrix from §2 again, read as two equations). For $x_1(0) = 3$, $x_2(0) = 0$ that's $f(0) = g(0) = 3/2$: the same amplitudes, and the same solution (2.11). Physically, f is the midpoint of the two carts and g is half the stretch of the middle spring, which is a nice way to see why the slow mode never stretches the middle spring and the fast one stretches it twice as much as either cart moves.

4 Why is the matrix diagonal in the eigenvector basis?

Felder and Felder say "This fact comes very directly from the definition of an eigenvector" and leave it there (p. 339; it's their Problem 6.206). We wrote it out under key point 1 in §3; here it is again, more slowly. The only two facts we need are that matrix multiplication is linear and the definition $\mathbf{T}\vec{v}_i = \lambda_i \vec{v}_i$.

Let $\vec{v}_1$, $\vec{v}_2$ be the eigenvectors of some transformation $\mathbf{T}$, with eigenvalues λ_1, λ_2. Take any vector and write it in the eigenvector basis:

$$\vec{v} = a\vec{v}_1 + b\vec{v}_2, \tag{4.1}$$

so $\vec{v}$ has coordinates (a, b) in the $\vec{v}_1\vec{v}_2$ basis. Apply $\mathbf{T}$. It acts on each piece separately (linearity), and by the definition of an eigenvector it does nothing to each piece except scale it:

$$\mathbf{T}\vec{v} = \mathbf{T}(a\vec{v}_1 + b\vec{v}_2) = a\,\mathbf{T}\vec{v}_1 + b\,\mathbf{T}\vec{v}_2 = a\lambda_1\vec{v}_1 + b\lambda_2\vec{v}_2. \tag{4.2}$$

Now read that in the $\vec{v}_1\vec{v}_2$ basis. The input had coordinates $\left(\begin{smallmatrix} a \\ b \end{smallmatrix}\right)$. The output has coordinates $\left(\begin{smallmatrix} \lambda_1 a \\ \lambda_2 b \end{smallmatrix}\right)$. The only matrix that takes (a, b) to $(\lambda_1 a, \lambda_2 b)$ is

$$\boxed{\mathbf{T} \text{ in the } \vec{v}_1\vec{v}_2 \text{ basis} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} \tag{4.3}$$

That's the whole proof. In the $\hat{\imath}\hat{\jmath}$ basis, $\mathbf{T}$ mixes the components (the off-diagonal entries are nonzero), because $\hat{\imath}$ and $\hat{\jmath}$ are not eigenvectors. In the eigenvector basis there is no mixing left to describe. Said another way: the columns of a matrix are the images of the basis vectors, and the image of $\vec{v}_1$ is $\lambda_1\vec{v}_1$, which in this basis is $(\lambda_1, 0)$.

For the springs, replace $\mathbf{T}$ by $\mathbf{M}$ and $\vec{v}$ by the state: plug (3.1) into (1.3) to get $\ddot{f}\vec{v}_1 + \ddot{g}\vec{v}_2 = f\lambda_1\vec{v}_1 + g\lambda_2\vec{v}_2$, and since $\vec{v}_1$ and $\vec{v}_2$ are independent, match coefficients: $\ddot{f} = \lambda_1 f$, $\ddot{g} = \lambda_2 g$. That is (3.3).

5 Which way should I use?

For one 2×2 problem by hand: it doesn't matter. The expensive part, finding the eigenvalues and eigenvectors (the determinant, the quadratic, two little linear systems), is shared, and neither method avoids it. After that, the first way substitutes each eigenvector in and fits the initial conditions with one 2×2 solve; the second way gets the decoupled equations in one line and fits the initial conditions with the same 2×2 solve dressed up as an inverse. The first way is a shade quicker (no inverse to write, and the substitution doubles as a check that your eigenvector is right); the second is a shade cleaner. Call it a wash.

The second way wins once the problem is bigger or gets reused:

- **Many degrees of freedom.** For N carts, the first way means N substitutions and an $N \times N$ fit; the second means one diagonalization and N independent one-dimensional oscillators. On a computer this is the whole story: `numpy.linalg.eig` once, then $\vec{x}(t) = \mathbf{P}\,\mathrm{diag}(\cos\omega_i t)\,\mathbf{P}^{-1}\vec{x}(0)$, one line for any N.
- **Reuse.** New initial conditions, a driving force, or damping proportional to velocity all leave the eigenvectors alone. The second way only redoes the two one-dimensional problems; the first way starts over from the substitution.
- **Understanding.** The second way is the one that says *why* only those two ratios work: only in the basis of the normal modes does the matrix become diagonal, and no other basis decouples the equations.

Felder and Felder's own framing (p. 340) matches this. In both approaches the takeaway is "the eigenvectors of the matrix define the normal modes of the system." The first way is how you compute a given problem; the second is why the computation works. "If you find the second approach confusing start by getting comfortable with the first approach ... but once you have worked with it a while come back to the second approach and see if you can see how it is doing the same thing as the first one, but in a different basis."

6 The general procedure, and the carts with unequal masses

1. Draw the picture.
2. Write down N2L for each cart.
3. Turn it into a matrix equation $\ddot{\vec{x}} = \mathbf{M}\vec{x}$.
4. Find the normal modes: the eigenvectors of $\mathbf{M}$ give the ratio of x_2 to x_1 in each mode, and the eigenvalues give $-\omega^2$.
5. Combine them, either way, and use the initial conditions.

For different masses m_1 and m_2 (still three equal springs), step 2 gives $m_1\ddot{x}_1 = -2kx_1 + kx_2$ and $m_2\ddot{x}_2 = kx_1 - 2kx_2$. Divide each line by its own mass and call the mass ratio $\mu = m_1/m_2$, so $k/m_2 = \mu\,(k/m_1)$:

$$\ddot{x}_2 = \mu\frac{k}{m_1}x_1 - 2\mu\frac{k}{m_1}x_2 \qquad \Rightarrow \qquad \mathbf{M} = \frac{k}{m_1}\begin{pmatrix} -2 & 1 \\ \mu & -2\mu \end{pmatrix}. \tag{6.1}$$

Plug in your μ and run step 4. The eigenvectors are no longer $(1, \pm 1)$, the heavier cart moves more in the slow mode and less in the fast one, and the ratio of the two periods is $T_{\text{slow}}/T_{\text{fast}} = \sqrt{\lambda_{\text{fast}}/\lambda_{\text{slow}}}$, which needs no k.

References

[1] Gary N. Felder and Kenny M. Felder, *Mathematical Methods in Engineering and Physics*, Wiley, 2016. Chapter 6, Sections 6.4, 6.8 and 6.9.